MIL-D-36256A
22 June 1976
SUPERSEDING
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5 August 1964

MILITARY SPECIFICATION

DISK, ABRASIVE, PLASTIC, DENTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plastic abrasive disks suitable for use during dental polishing procedures.

1.2 Classification. Plastic abrasive disks shall be of the following types and grit sizes as specified (see 6.1).

Type I. - Flint (Quartz or Quartzite)
Grit sizes: - 150, 220, 280

Type II. - Aluminum Oxide
Grit sizes: - 120, 220

Type III. - Silicon Carbide
Grit sizes: - 80, 120, 220

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Attn: Directorate of Medical Materiel, CODE ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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SPECIFICATIONS

FEDERAL

L-P-377	- Plastic Sheet and Strip, Polyester.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D882	- Tests for Tensile Properties of This Plastic Sheeting

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

U.S. DEPARTMENT OF COMMERCE

NATIONAL BUREAU OF STANDARDS

Product Standard PS8 - Grading of Abrasive Grain on Coated Abrasive Products

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(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20234.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive. The abrasive shall be of crushed materials and crystalline in form. The abrasive shall be clean, of high quality, and free from extraneous materials. Type I abrasive grit shall be flint (quartz or quartzite) free from extraneous materials. Type II abrasive grit shall be aluminum oxide, free from extraneous materials. Type III abrasive grit shall be crushed silicon carbide, of high quality and free from extraneous materials.

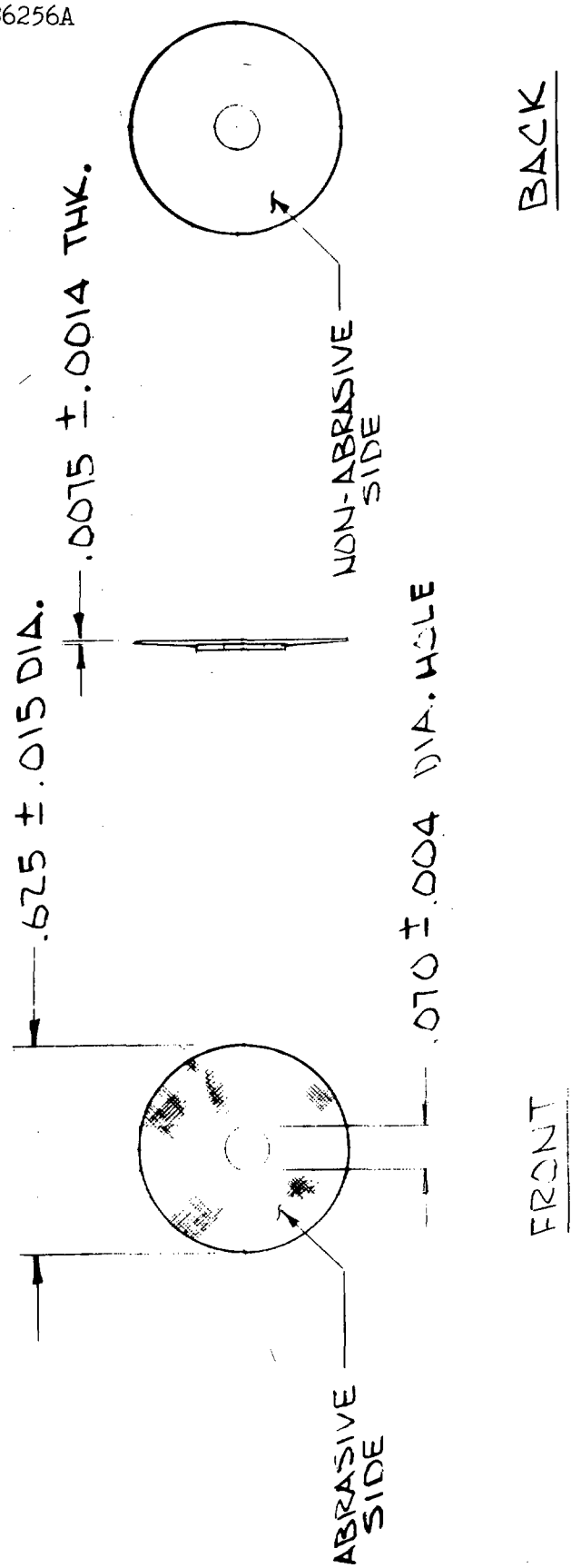
3.1.1.1 Grading. Grit sizes shall conform to 1.2 for the applicable type (flint, aluminum oxide or silicon carbide) specified (see 6.1). Grit sizes 80, 120, 150, and 220 shall conform to product standard PS 8 for screen grit sizes (see 6.3). The 280-grit flint shall conform to product standard PS 8 for sedimentation grit sizes (see 6.3).

3.1.2 Bonding agent. The bonding agent shall be a high quality waterproof, resin-type bonding agent suitable for the purpose intended.

3.1.3 Backing. The backing shall be polyethylene terephthalate film (polyester) without the addition of dyes or pigments. Polyethylene terephthalate film shall be in accordance with the applicable requirements of type I, general purpose film of L-P-377, except that the haze requirement shall not apply. Tests for tensile strength and elongation shall be conducted in accordance with 4.4.2 of this specification.

3.2 Design and construction. The disks shall be flat, safe-sided, plastic abrasive disks suitable for dental polishing procedures in the oral cavity. The disk shall be in accordance with figure 1 for design and dimensions and shall have a hole concentric within 0.005 inch making it suitable for mounting and use with either a straight handpiece or an angle handpiece No. 303 screwhead dental mandrel. One side of the disk shall be coated with abrasive grain of the applicable type and grit size specified in 1.2 and 6.1 for the applicable type and grit size. The abrasive coating shall be of the closed coat type and the abrasive shall be bonded in such a manner as to uniformly and suitably cover the backing surface to produce a sharp-cutting abrasive product. The abrasive shall wear off gradually when in use. The other side of the backing shall be uncoated. Unless otherwise specified, a minimum of 100 disks shall be furnished in each unit package (see 6.1).

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DISK, ABRASIVE, PLASTIC, DENTAL

FIGURE 1

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3.3 Performance. The abrasive disks shall perform satisfactorily both dry and wet, even after a one-hour soak in distilled water. The disks shall run true and wear evenly, and the binder shall not fuse, glaze, or create heat when in use. The disks shall be pliable yet possess sufficient stiffness. The disks shall stay flat without warping under normal operative procedures, even after an oven heat treatment at 180 degrees Fahrenheit (°F.) for 50 minutes when tested as specified in 4.4.3.

3.4 Workmanship. Workmanship shall be first class throughout. Disks shall be free from defects which detract from their appearance or may impair their serviceability. Disks shall be flat, safe-sided and cleanly cut.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the abrasive, bonding agent, and backing material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, III and IV (see 4.2.1, 4.3 and 4.3.1) and table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be the individual disk.

TABLE I. Sampling for examination

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be the individual disk.

TABLE II. Sampling for tests

Component	Characteristic	Requirement	Test procedure	Unit of product	Inspection level
Raw abrasive	Grading	3.1.1	4.4.1	N/A	1/
Uncoated backing	Tensile strength and elongation	3.1.3	4.4.2	N/A	1/
Finished item	Performance	3.3	4.4.3	1 disk	S-1

1/ Test may be performed either on raw material or on end item (by recovery procedures). When testing is done on raw material, sufficient amount from each lot of applicable component shall be taken for testing. When testing is done on end item, sufficient amount of applicable component shall be taken from each lot of end item by recovery procedures to perform applicable test.

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4.3 Examination. The disks shall be examined for defects including, but not limited to, those listed in table III.

TABLE III. Classification of defects

Categories	Defects
Major A:	
101	Type and grit size of disk not as specified.
102	Disks not symmetrically made.
103	Disks not flat.
104	Holes not present; not centered.
105	Disks not uniformly covered with abrasive on one side.
106	Disks not safe sided.
Major A:	
107	Cracks, chips, holes, cuts, tears, or roughly cut edge present in disks.
108	Foreign material present in disk abrasive.
Major B:	
151	Thickness of disks not as specified.
152	Diameter of mandrel hole not within specified limits.
153	Size (diameter of disk) not as specified.
154	Count not as specified.
Minor:	
201	Foreign material present on safe side of disks.
202	Creases present in disks.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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TABLE IV. Examination of preparation for delivery

Examination	Defects
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.4 Tests.

4.4.1 Abrasive grading. The grading of the abrasive grit shall be tested in conformance with product standard PS 8.

4.4.2 Backing. The tensile strength and elongation of the backing (prior to coating) shall be tested in accordance with method A of ASTM D882. The specimen size shall be 1 inch wide by 5 inches long. Elongation rate shall be one hundred percent per minute. Separation between jaws shall be 2 inches. Specimen thickness shall be determined before testing.

4.4.3 Performance test. Disks shall be mounted in a No. 303 dental mandrel inserted in a dental handpiece of a conventional dental engine and used for smoothing and polishing dental cement and plastic restorations both under wet and dry conditions. The performance shall be evaluated for properties specified in 3.3.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One box containing 100 disks, as specified, constitutes one unit of issue.

5.1.1.1 Level A.

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5.1.1.1.1 Unit package. Each unit of issue shall be packaged in a box of appropriate size and design. Closure shall be adequate to prevent accidental opening.

5.1.1.1.2 Intermediate package. Twelve (12) units of issue shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Level C. Units of issue shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. One thousand six hundred and twenty (1620) units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 of PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. One thousand six hundred and twenty (1620) units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Units of issue shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit and intermediate packages, and exterior containers. Unit and intermediate packages and exterior containers shall be marked as specified in MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and grit size required (see 1.2).
c. If unit of issue is different from that specified (see 3.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

Specification Designation			
Type	Grit Size	Stock Number	Item Identification
I	150	6520-00-226-0571	DISK, ABRASIVE, Flint, 150 Grit, Plastic, 5/8 inch Diameter, 100s
I	220	6520-00-226-0572	DISK, ABRASIVE, Flint, 220 Grit, Plastic, 5/8 inch Diameter, 100s
I	280	6520-00-226-0573	DISK, ABRASIVE, Flint, 280 Grit, Plastic, 5/8 inch Diameter, 100s
II	120	6520-00-226-0574	DISK, ABRASIVE, Aluminum Oxide, 120 Grit, Plastic, 5/8 inch Diameter, 100s
II	220	6520-00-226-0575	DISK, ABRASIVE, Aluminum Oxide, 200 Grit, Plastic, 5/8 inch Diameter, 100s
III	80	6520-00-226-0576	DISK, ABRASIVE, Silicon Carbide, 80 Grit, Plastic, 5/8 inch Diameter, 100s
III	120	6520-00-226-0577	DISK, ABRASIVE, Silicon Carbide, 120 Grit, Plastic, 5/8 inch Diameter, 100s
III	220	6520-00-226-0578	DISK, ABRASIVE, Silicon Carbide, 220 Grit, Plastic, 5/8 inch Diameter, 100s

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6.3 Grit sizes. Screen grit sizes are those which are coarse enough to be measured and controlled by use of sieves. Sedimentation grit sizes are those which are so fine that they have to be measured and controlled by a sedimentation method based on Stokes Law.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1405

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-D-36256A - DISK, ABRASIVE, PLASTIC, DENTAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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